Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Soller, Babs R. Ph.D.		
Project Title:	Noninvasive Measurement of Blood and Tissue Chemistry		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Previous affiliation was University of Massachusetts Medical School, Worcester, MA, until October 2010 (Ed., 11-9-2010)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heard, Stephen (University of Massachusetts Medical School)		
Grant/Contract No.:	NCC 9-58-SMS00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Trauma and acute medical problems along with loss of skeletal muscle mass, strength and endurance, are some of the most serious risks facing astronauts during long duration spaceflight. The measurement of two metabolic parameters, muscle pH and oxygen tension (PO2) can be applied in both areas. This project examines the hypothesis that near infrared spectroscopy (NIRS) is a platform technology that can provide noninvasive physiologic monitoring in support of multiple NASA needs. The immediate goal is to produce, validate and deliver to NASA-JSC a small, lightweight medical monitor which utilizes NIRS to measure important metabolic parameters. Requisite precision and accuracy will be demonstrated for both male and female subjects of any ethnic origin. The device will perform equally well on the forearm and the thigh, despite gender and weight related differences in fat thickness. The specific aims of this project are: (1) Develop calibration procedures for modified NIRS sensor to measure deep muscle metabolic parameters (tissue pH and PO2); (2) Validate sensor in exercise protocol and deliver systems to NASA-JSC; (3) Determine values for tissue pH and PO2 that can be used by a smart medical system to indicate shock and assist in guiding treatment; and (4) Evaluate hardware for flight requirements and develop plan to produce flight-ready instrumentation. During year one we developed calibration equations to relate near infrared spectra collected transdermally from the forearm of humans subjects to muscle pH, muscle PO2 and blood hematocrit. We showed that these calibration equations had the required accuracy, compared to an invasive sensor, for the intended clinical applications. We made these measurements on the forearm muscle whose strength is most important for successful completion of an extravehicular activity (EVA). We demonstrated that muscle pH is a sensitive measure of work intensity, more sensitive than blood measurements of pH or lactate. With Dr. Hagan, Manager of JSC's Exercise Physiology Lab, we are exploring the use of the noninvasive monitor to assess astronaut fitness and readiness for EVA in space. Also this year we developed a set of tools to help us optimize sensor and system design to be able to accurately measure the metabolic parameters on the thigh muscle, which is covered with a thick layer of fat. One of these tools is a mathematical model which describes the photon path through skin and fat to the muscle. This model was validated with a prototype sensor which had flexible source-detector spacing and a new type of tissue phantom based on a stable NIR dye which mimics deoxyhemoglobin absorbance. Sensor specifications were determined and fat correction was demonstrated on the phantom materials. Initial human studies showed that the technique is effective in removing the spectral influence of fat. System modifications are underway to improve the monitor for adequate light detection from muscle beneath thick fat layers. In parallel, the human protocol has been redesigned to validate sensors using a leg extension model. This change was approved by NASA-JSC and will allow us to use NASA equipment for the study and validate our results against other studies in the literature. During the next year of the project we will complete validation of the sensor on the thigh, and deliver this system to NASA-JSC for evaluation during cycle ergometry in their laboratory. We conduct a weekly telecon with Dr. Hagan and have been visiting once/month. Once we have systems on-site at JSC (spring 2005) a graduate student will spend more time at JSC to support integration of the system into the exercise physiology laboratory. Additionally, we will ramp up our interaction with JSC engineering to begin to address equipment issues related to flight. The other application of the system will be in a Smart Medical System for the assessment and treatment of trauma or injury which results in reduced blood pressure, blood flow or cardiac output. Values for muscle pH and PO2 that can be used to assess severity of injury and guide treatment will be determined from a study of patients who enter the emergency room in shock. This study is expected to begin at UMass in early fall, 2005. The NIRS noninvasive metabolic monitor is expected to have many applications for NASA. The system will have additional use on earth for military and civilian personnel treating mass casualties. It can also be used in the hospital, ambulances and helicopters. As part of a Smart Medical System, advanced medical assessment and monitoring may become available to physicians in remote and rural areas, who may not have access to specialist expertise. The monitor is expandable to measure new chemistries by just altering the calibration equations stored in computer memory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will have direct earth based application. The prototype monitors we are developing will have application in emergency response vehicles, emergency rooms, and hospitals. Pre-hospital applications include assessing the severity of shock and triaging multiple casualties, as well as providing a sensor for a smart medical system to guide resuscitation from hemorrhage. In the ICU we expect that this monitor will find application in distinguishing between hemorrhagic and septic shock. The direct muscle application of interest to NASA for assessing fitness in space, may be useful to assess success of physical therapy in rehabilitating patients with muscle injury or atrophy. There is also general medical application for the diagnosis of anemia and if small and inexpensive enough, screening world-wide for malnutrition. There is also a possible application for the diagnosis of diabetic foot ulcers. A smaller version of this monitor could find use in the training of elite and weekend athletes.
Task Progress:	We developed a mathematical model of spectra for human tissue which takes into account the skin, fat and muscle layers. The model was used, in conjunction with tissue mimicking phantoms, to better understand the influence that fat thickness and its absorbance and scattering properties have on the absorption spectrum used to calculate muscle pH, muscle PO2 and blood hematocrit. We created a novel tissue mimicking phantom based on a near infrared dye with absorption properties similar to deoxyhemoglobin. Measurement on these phantoms was used to demonstrate prediction capabilities of the mathematical (Monte Carlo) model. The phantoms and Monte Carlo model helped us identify key sensor design issues to assure optical depth penetration through fat on the forearm and the thigh. A flexible test probe was designed that allowed us to investigate a number of features such as angle and source-detector spacing. A new fiber optic probe, different from the one previously used on the palm, was implemented for exercise studies on the arm. Previously we had used cardiac surgery patients as a subject group to vary pH, PO2 and hematocrit. However, we found that we could not consistently achieve a wide range of pH and PO2 to derive robust calibration equations. We instituted a repetitive handgrip exercise protocol and began monitoring on the forearm. This protocol allows us to compare our measurements with blood and intracellular measurements obtained by other investigators. We were also able to demonstrate the NIRS technique on moving muscle, for the first time. We completed 34 subjects, which has allowed us to develop calibration equations to calculate muscle pH and PO2 over a wide range. We are currently validating these equations on an independent group of subjects.

	During the course of this study we were also able to investigate the relationship between muscle pH, PO2, PCO2 and blood values. We determined that the muscle values are more sensitive than the blood values to increases in exercise intensity and have the potential to be used as a noninvasive measurement to assess metabolic status of the muscle and the effectiveness of exercise countermeasures. Also in this year we designed and built a prototype of a unique fiber optic probe to measure thigh muscle through thick layers of fat. Using novel mathematical methods, which we developed, we were able to demonstrate improvement in the accuracy of hemoglobin measurement on phantoms. In preliminary experiments we were able to show that the sensor was effective in correcting the spectra of anatomical features with different fat thicknesses (forearm, upper arm, calf and thigh) so they all look the same, irrespective of fat thickness. Finally, we developed a human testing protocol for evaluating exercise on the thigh using both static and dynamic knee extension exercise. NASA-JSC will loan us their exercise equipment which is has been used to assess muscle strength before and after spaceflight.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Articles in Peer-reviewed Journals	Soyemi, O. O., Landry, M. R., Yang, Y., Idwasi, P. O., Soller, B. R. "Skin Color Correction for Tissue Spectroscopy: Demonstration of a Novel Approach with Tissue Mimicking Phantoms Applied Spectroscopy" 237-244 , Feb-2005
Articles in Peer-reviewed Journals	Yang, Y., Soyemi, O. O., Landry, M. R., Soller, B. R. "Influence of a fat layer on the near infrared spectra of human muscle: quantitative analysis based on two-layered Monte Carlo simulations and phantom experiments. Optics Express" Jan-2005
Awards	" Associate Team Leader, Smart Medical Systems Team, NSBRI " Jan-2005
Awards	" Selected for US Army Medical Research Command Pilot Program in Technology Commercialization " Jan-2005
Patents	Soyemi, O. O. and Soller, B. R. "Removal of Skin Pigmentation from Spectra. " Jan-2005
Presentation	Soller, B. R. "Near Infrared Spectroscopy to Assess Metabolic Parameters to Guide Resuscitation from Shock. " Apr-2004
Presentation	Soller, B. R. "New Means of Assessing Tissue Oxygenation at the Bedside. " Jan-2005
Presentation	Soller, B. R. "Noninvasive Measurement of Blood and Tissue Chemistry. " Jan-2005
Presentation	Soller, B. R. "Noninvasive Monitoring of Tissue Oxygenation Parameters (pH, PO2, Hct) Using Near Infrared Spectroscopy. Applications in Shock. " Aug-2004
Presentation	Soller, B. R. and Hagan, R. D. "Noninvasive Sensor for the Measurement of Muscle Metabolism During Exercise " Jan-2004
Presentation	Soller, B. R. and Soyemi, O. O. "Noninvasive Monitor for Continuous Measurement of Blood Hematocrit, Muscle pH and PO2. " Oct-2004
Presentation	Soller, B. R., and Phillipps, P. "Portable, Noninvasive System for Use in the Triage and Treatment of Shock: System Design and Characterization. " Apr-2004
Presentation	Soyemi, O. O. "Removal of the spectral effects of skin pigmentation in spectra: Demonstration with tissue mimicking phantoms. " Oct-2004